Lab 5: Static Code Analysis
Outputs of Pylint, Bandit and Flake8

Project: Bike Share Stations | File: bike_share_stations.py

Evidence compiled from the terminal reports provided during the lab on 16 October 2026. Original findings and final scan outputs are shown below.

Analysis tool	Before fixing	After fixing
Pylint	3.89 / 10	10.00 / 10
Bandit	3 findings (1 medium, 2 low)	No issues identified
Flake8	10 violations	No violations

1. Pylint - Original output

Initial rating: 3.89/10. Original warnings reported:

```
5:0   C0301  Line too long (106/100)
9:0   C0301  Line too long (131/100)
27:27 C0303  Trailing whitespace
1:0   C0114  Missing module docstring
8:0   C0116  Missing function/method docstring
8:0   C0103  Function name addStation not snake_case
8:0   W0102  Dangerous default value [] as argument
14:0  C0116  Missing function/method docstring
14:0  C0103  Function name rentBike not snake_case
14:0  R1710  Inconsistent return statements
21:0  C0116  Missing function/method docstring
21:0  C0103  Function name unlockPin not snake_case
25:0  C0116  Missing function/method docstring
25:0  C0103  Function name fullStations not snake_case
27:4  W0612  Unused variable 'summary'
31:0  C0116  Missing function/method docstring
31:0  C0103  Function name resetStation not snake_case
34:11 W0718  Catching too general exception Exception
38:0  C0116  Missing function/method docstring
38:0  C0103  Function name hasBikes not snake_case
40:7  C0121  Use is None instead of == None
45:0  C0116  Missing function/method docstring

Your code has been rated at 3.89/10
```

Pylint - Final output

```
Your code has been rated at 10.00/10
(previous run: 3.89/10, +6.11)
```

2. Bandit - Original output

Original scan detected 3 potential issues: one medium severity and two low severity.

```
Issue: B608  hardcoded_sql_expressions
Location: bike_share_stations.py:9:17
Severity: Medium | Confidence: Medium
Possible SQL injection through string-based query construction.

Issue: B311  blacklist
Location: bike_share_stations.py:22:11
Severity: Low | Confidence: High
Pseudo-random generator used for a security-sensitive PIN.

Issue: B110  try_except_pass
Location: bike_share_stations.py:34:4
Severity: Low | Confidence: High
Try, Except, Pass detected.

Original Bandit run metrics:
Total lines of code: 36
Issues by severity: Undefined 0 | Low 2 | Medium 1 | High 0
Issues by confidence: Low 0 | Medium 1 | High 2
```

Bandit - Final output

```
Test results:
    No issues identified.

Code scanned:
    Total lines of code: 84
    Total lines skipped (#nosec): 0
    Total potential issues skipped: 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
Files skipped (0):
```

3. Flake8 - Original output

```
bike_share_stations.py:5:80: E501 line too long (106 > 79)
bike_share_stations.py:9:80: E501 line too long (131 > 79)
bike_share_stations.py:15:80: E501 line too long (91 > 79)
bike_share_stations.py:17:80: E501 line too long (89 > 79)
bike_share_stations.py:26:80: E501 line too long (90 > 79)
bike_share_stations.py:27:5: F841 unused variable 'summary'
bike_share_stations.py:27:28: W291 trailing whitespace
bike_share_stations.py:33:80: E501 line too long (81 > 79)
bike_share_stations.py:39:80: E501 line too long (91 > 79)
bike_share_stations.py:40:15: E711 comparison to None
```

Flake8 - Final output

```
$ flake8 bike_share_stations.py
(no output - no violations)
$ flake8 bike_share_stations.py && echo "Flake8: No violations found"
Flake8: No violations found
```

Conclusion: The revised Python file achieved Pylint 10.00/10, zero Bandit findings and zero Flake8 violations. These are static-analysis outcomes, not proof that every possible runtime issue is absent.